

B4M36SAN
BE4M36SAN
Seminar/Tutorial

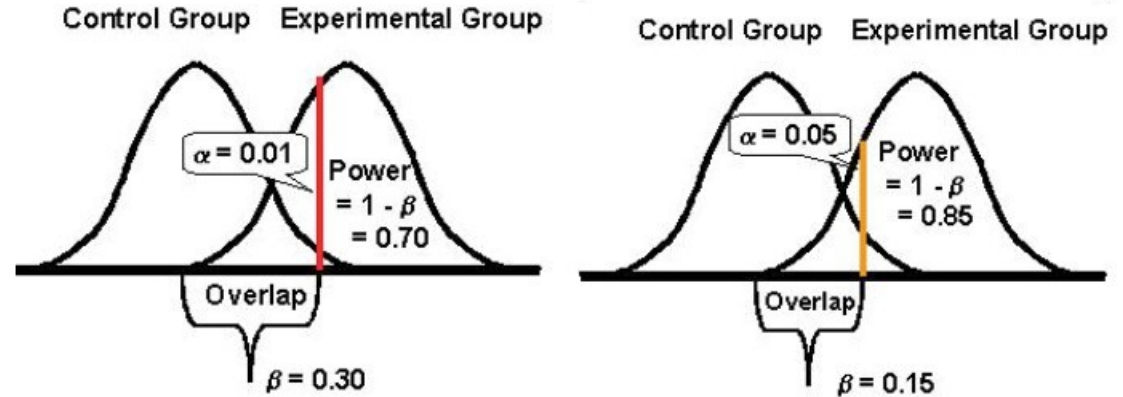
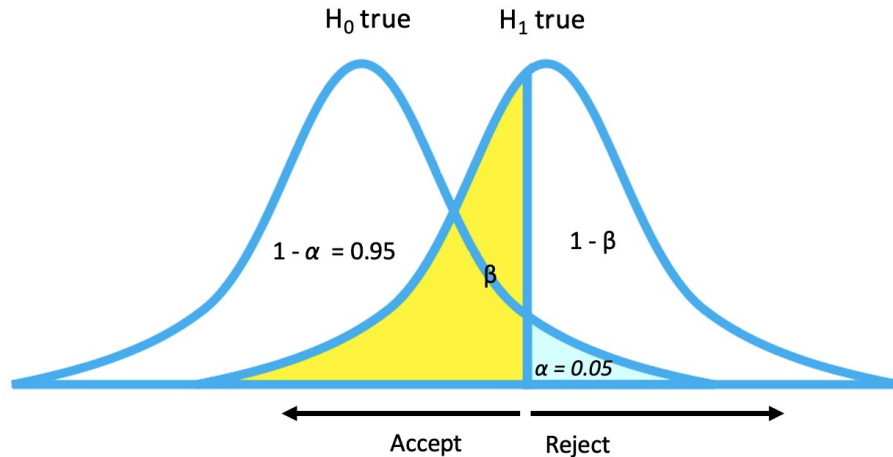
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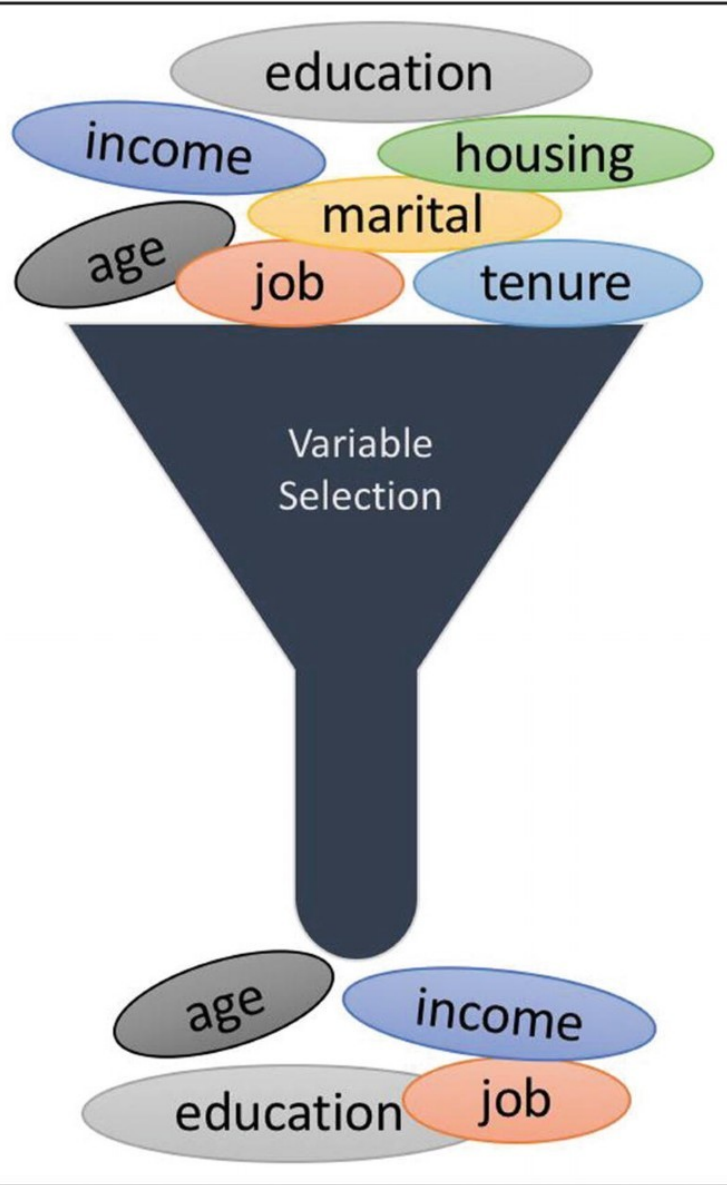
A recap of type1 – type 2 errors

By changing the “threshold” we can select needed ratio of Type 1 to Type 2 errors

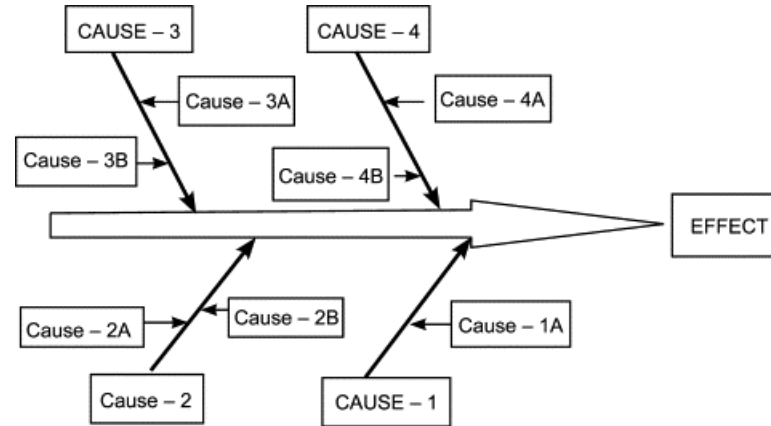


- To simplify the analysis we actually just fix one of errors, Type 1 = 0.05 = confidence
- After that we calculate the Type 2 and denote it as **power** of the test

<https://youtu.be/OhDZIO8vtGw?si=yK76iwA0wftf2w4A&t=400>

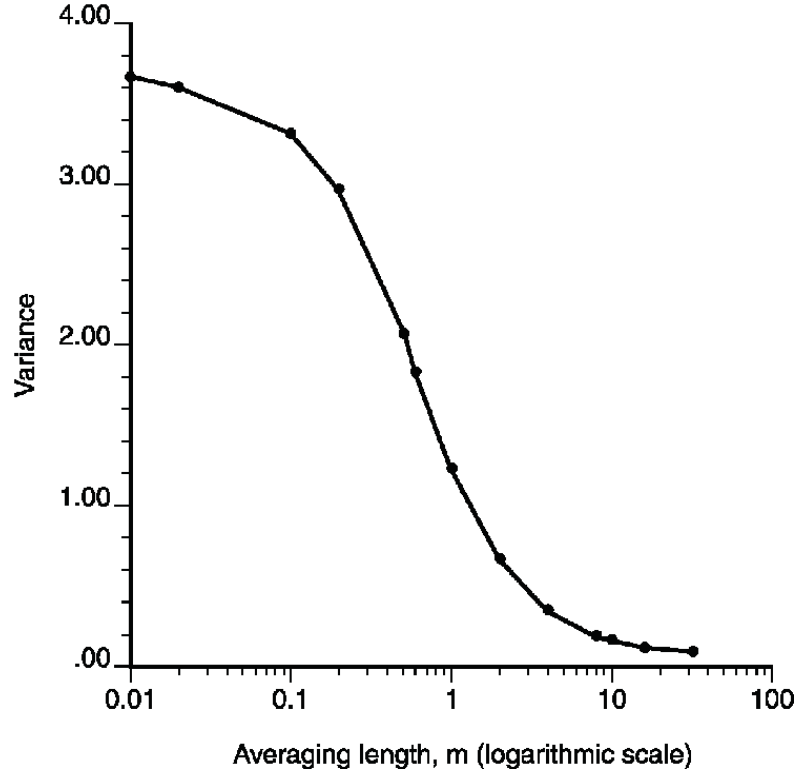


- Up until now the randomness was assumed to be as an absence of variables, which leads to non-100% calculations
- However most of the time even measured variables are TOO MUCH to practically compute
- Even though they DO affect the outcome, the size of the effect is too low

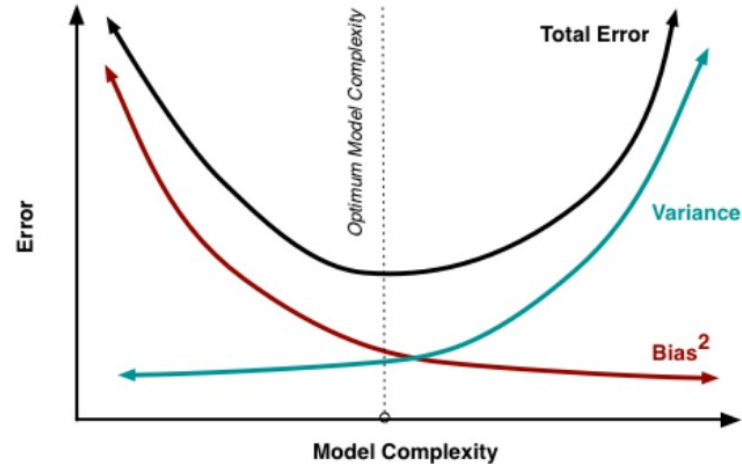


Cause-and-Effect Diagram

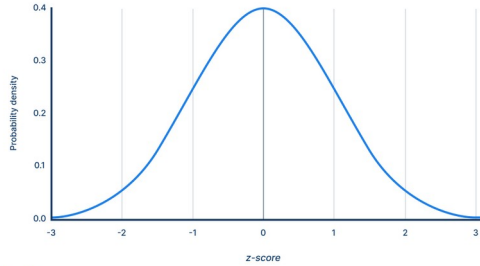
Variable selection



- You can think of this as following:
 - Addition of more variables makes the prediction more precise (smaller variance)
 - But decrease from 1.1% to 1.09% is too small and the computational complexity is spent for nothing
- Moreover, with more variables we need more and more samples to maintain same variance
 - Bias-variance tradeoff



Standard normal distribution



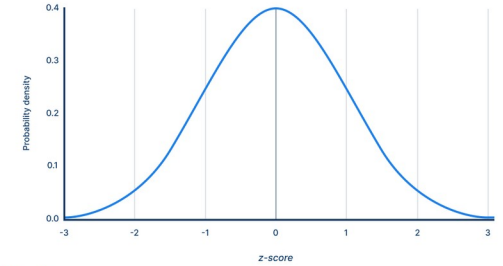
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Standard normal distribution



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Standard normal distribution

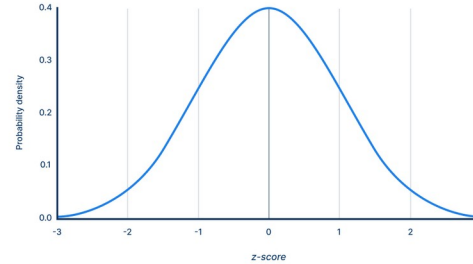


Standard normal distribution



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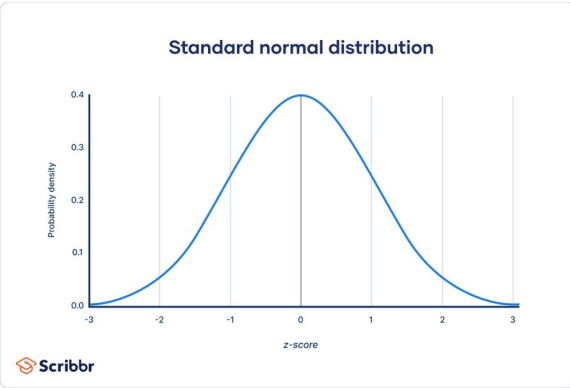
Standard normal distribution



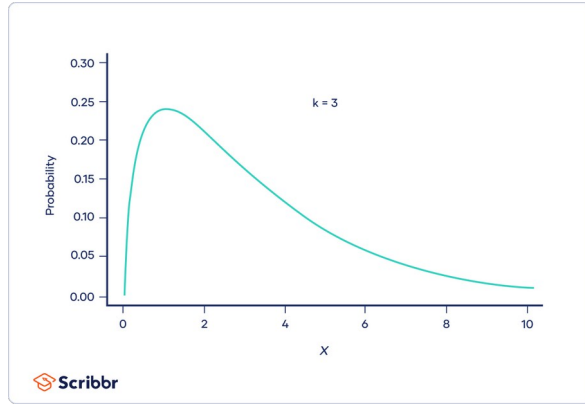
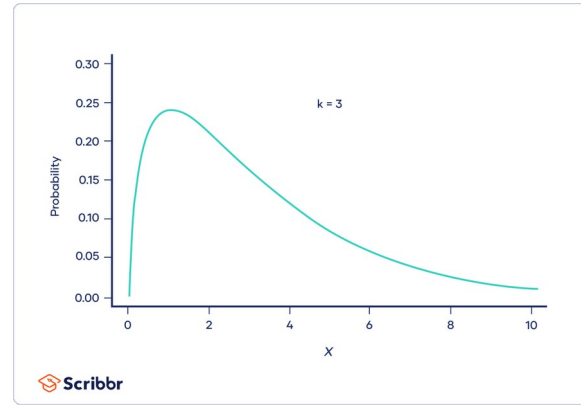
https://en.wikipedia.org/wiki/Sum_of_normally_distributed_random_variables

Distribution arithmetic: Normal distribution

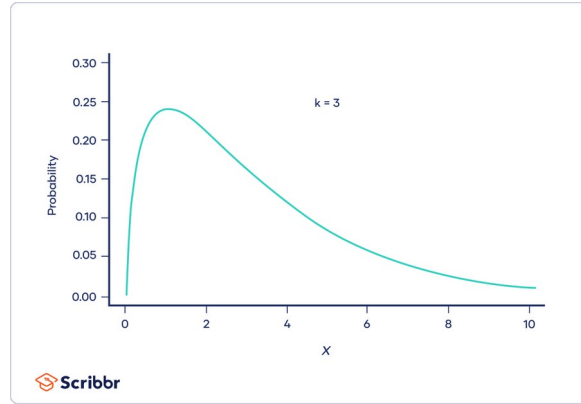
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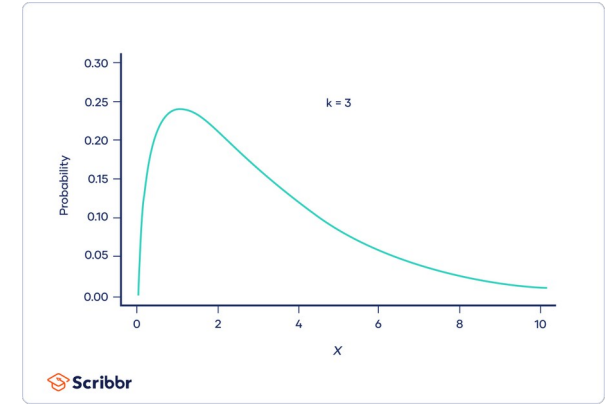
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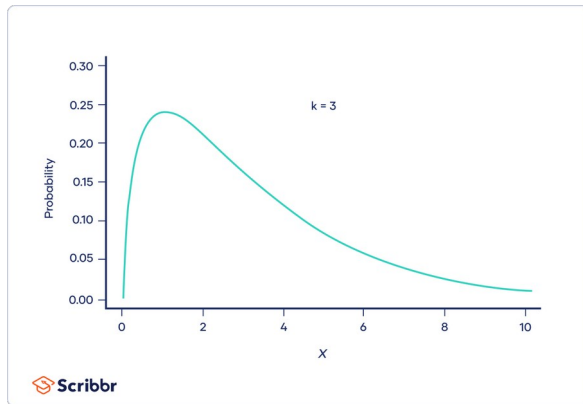
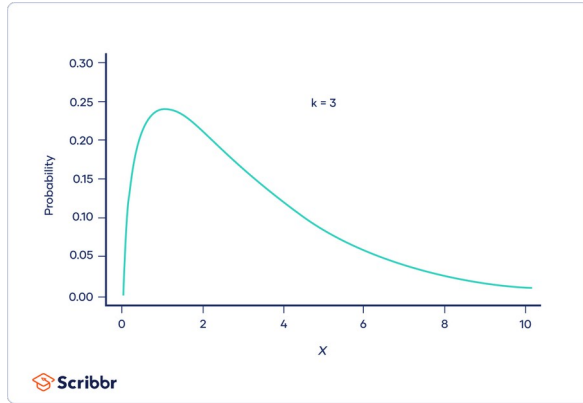
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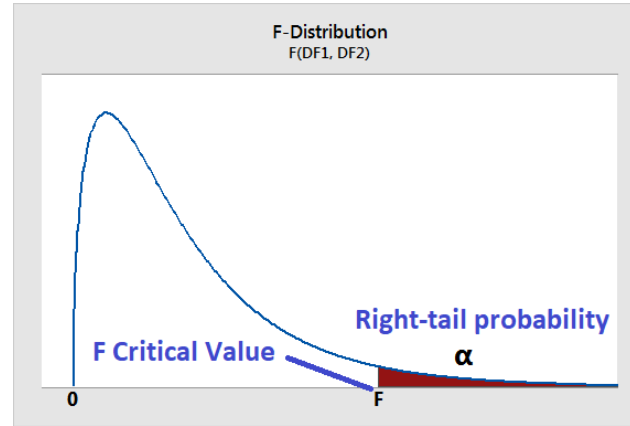
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Distribution arithmetic: Chi squared distribution



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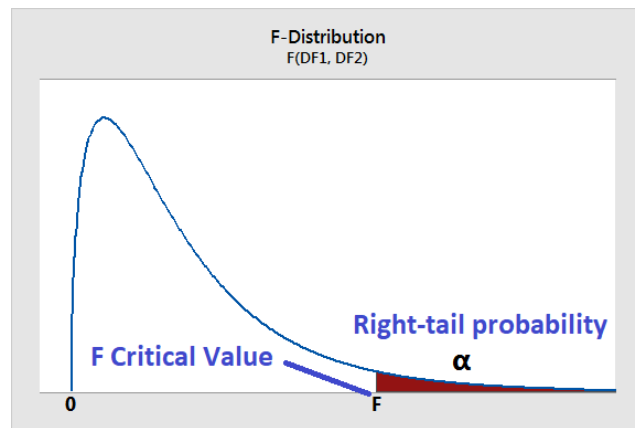


Distribution arithmetic: F-distribution = division of chi sq

$$Z_1^2 + Z_2^2 + Z_3^2 + \dots + Z_n^2$$

$$Z_1^2 + Z_2^2 + Z_3^2 + \dots + Z_n^2$$

=



The Population Variance Formula

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

x_i are the individual data values
 μ is the population mean
 N is the number of data values

Distribution arithmetic: F-distribution (division of variances)

Problem:

$$RSS = \sum_{\vec{x}} (y_i - x_1 \cdot \beta_1 - x_2 \cdot \beta_2 - \beta_0)^2 \rightarrow \text{local optimum (estimate): } \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_0$$

adjusted/normal $R^2 \gg 0$ (at least 0.5) AND $F\text{-stat } p\text{-value} < 0.05 \rightarrow H_A \text{ is true}$

H_A : X_1, X_2 and Y have some relationship for at least one : $\beta_1 \neq 0$ or $\beta_2 \neq 0$

But which of variables are exactly non-zero?

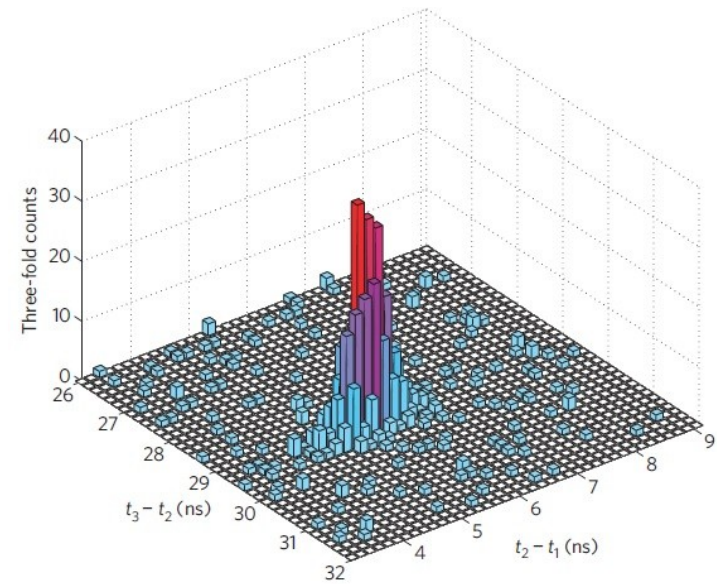
	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Recall:

$$\frac{\hat{\beta}_1 - 0}{S_{\beta_1}} \in t\text{-student}(df = m - 2)$$

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More variables -> Curse of dimensionality(a lot of empty cells) -> low S_{β_1} -> T – test is often passed randomly:

UNDERFIT!

In other words – due to the lack of sample numbers/large dimension/bad luck we get samples that unfortunately shows an illusion of dependency – so called **SELECTIVE BIAS**

Please open the today's activity .zip file and find the:

LREG_artificial_irrelevant.R

You will find the problem with t-test p-values in linear regression in first block

Artificial data

We will generate artificial data first. The dataset contains 100 independent variables and one dependent variable, all continuous. All the variables follow the normal distribution with zero mean and unit variance. Notice that there is **no relationship between dependent and independent variables**. Consequently, all further findings must be false positives (Type I error).

```
set.seed(3) # you can change the seed or remove the line to see more runs
d<-data.frame(matrix(rnorm(200*100,0,1),nrow=200))
d$out<-rnorm(200,0,1)
```

Learn a linear model

Next, we will create a linear regression model. There is no feature selection in this step. Check the number of seemingly relevant independent variables, their number should be around $\alpha \cdot nvars = 5$. You may need to generate data multiple times to get an "illustrative" case.

```
lm.all<-lm(out ~ .,data=d)
lm.all.par<-coefficients(summary(lm.all))
sum(lm.all.par[, "Pr(>|t|)"]<0.05)
```


Alternative solution = instead of looking at p-values, use F-test

Straightforward solution = just check all subsets

A little **heuristic** = greedily check until it increases and stop afterwards

$$TSS = \sum (y_i - \bar{y})^2 \quad RSE = \sqrt{\frac{1}{n-p-1} RSS},$$

$$RSS = \sum (y_i - \hat{y}_i)^2 \quad R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

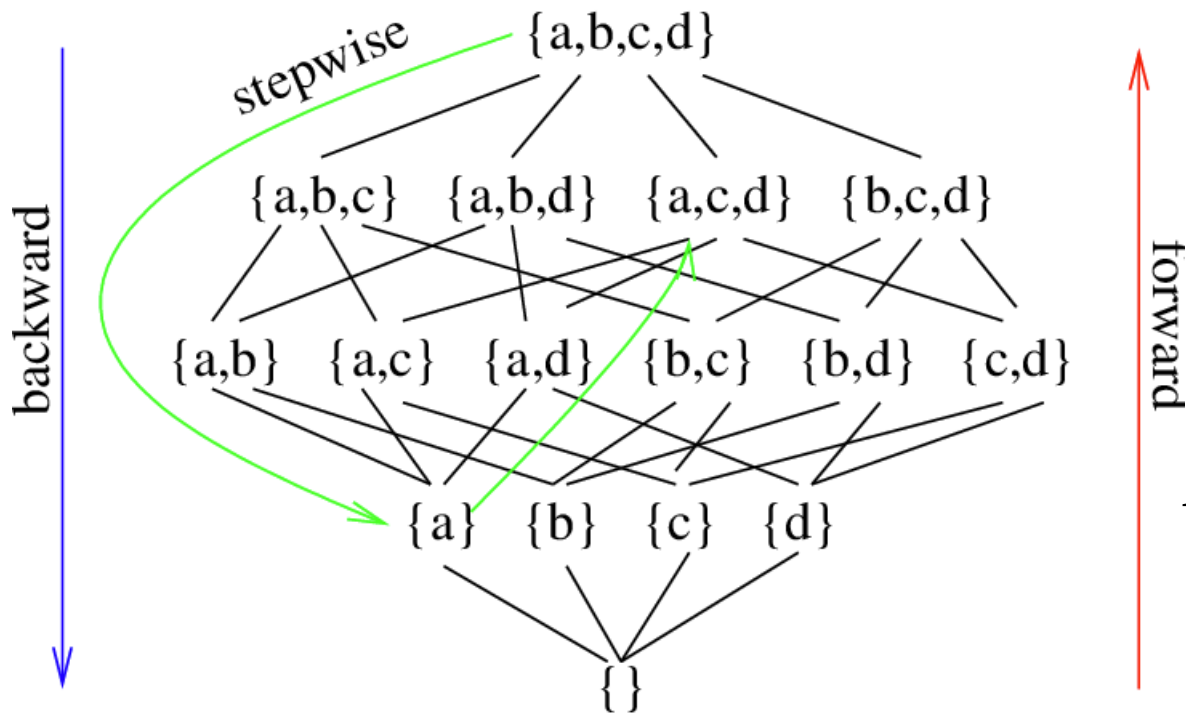
$$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)},$$

$$F\text{-stat} \propto \text{adjusted } R^2$$

Alternatively: AIC (same idea)

$$R^2 = 1 - RSS/TSS$$

$$AIC = \frac{1}{n\hat{\sigma}^2} (RSS + 2d\hat{\sigma}^2)$$



AIC vs adjusted R², what is the difference?

The key difference between adjusted R² and AIC/BIC:

- Adjusted R² is used only in simple linear regression
- AIC/BIC can be used for nearly all models/algorithms
 - We will show such advanced method in upcoming seminars
- They are very similar, but adjusted R² tends to choose more complex models

Control question:

Why are we using RSS instead of maximum likelihood in simple linear regression? Some hint:

$$AIC = -2 \ln(L) + 2k$$
$$R^2 = 1 - RSS/TSS$$
$$AIC = \frac{1}{n\hat{\sigma}^2} (RSS + 2d\hat{\sigma}^2)$$

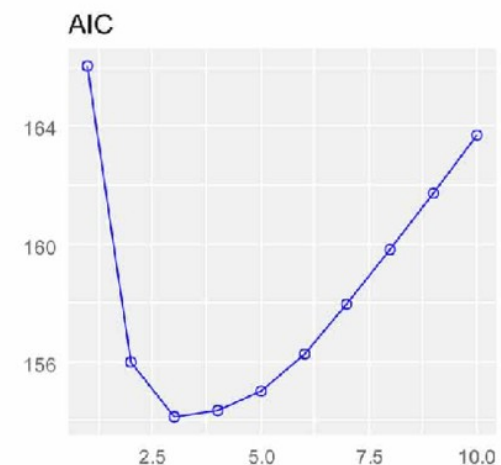
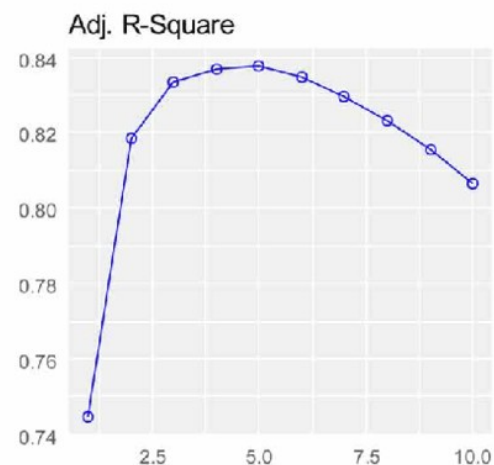
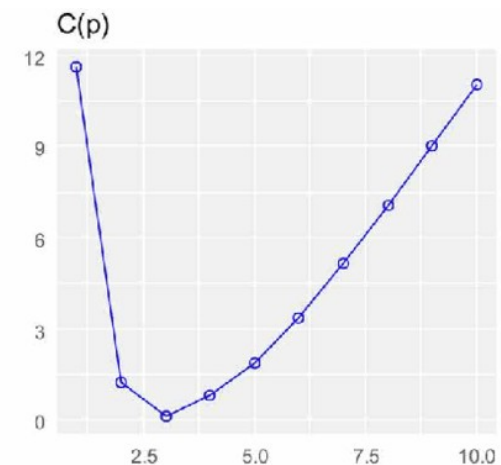
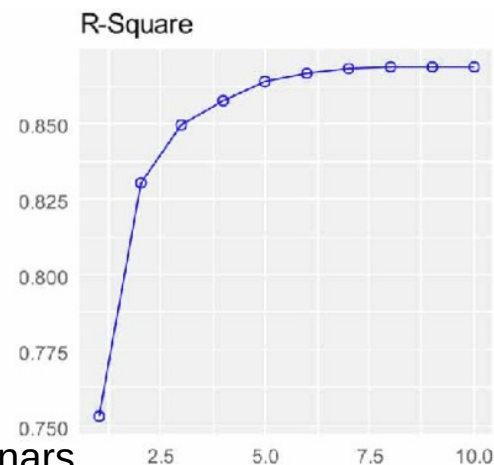


Figure 4

You can compare F-stats for stepwise neighbors:

$$F = \frac{s_1^2}{s_2^2}$$

BUT: Only if models are nested! E.g. second model is addition of some variables

Model1: $Y \sim X_1 + X_2$

Model2: $Y \sim X_1 + X_2 + X_3 + X_4 + \dots$

$$\text{where } s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$ANOVA(model_1, model_2) = \frac{F - stat_1}{F - stat_2} = \frac{S_1}{S_0} \div \frac{S_2}{S_0} = \frac{S_1}{S_2} = F - stat$$

Formally:

p-value < 0.05 -----> model2 is better! (adjusted R^2 model2 >> adjusted R^2 model1)

p-value > 0.05 -----> model1 is better! (adjusted R^2 model2 << adjusted R^2 model1)

In R:

$$\begin{aligned} &Y \sim X_1, \\ &Y \sim X_1 + X_2, \\ &Y \sim X_1 + X_2 + X_3, \dots \end{aligned} \qquad = anova(model_1, model_2), anova(model_2, model_3), \dots$$

But people realized that such method is too bad in practice

Here are *some* of the problems with stepwise variable selection.

1. It yields R-squared values that are badly biased to be high.
2. The F and chi-squared tests quoted next to each variable on the printout do not have the claimed distribution.
3. The method yields confidence intervals for effects and predicted values that are falsely narrow; see Altman and Andersen (1989).
4. It yields p -values that do not have the proper meaning, and the proper correction for them is a difficult problem.
5. It gives biased regression coefficients that need shrinkage (the coefficients for remaining variables are too large; see Tibshirani [1996]).
6. It has severe problems in the presence of collinearity.
7. It is based on methods (e.g., F tests for nested models) that were intended to be used to test prespecified hypotheses.
8. Increasing the sample size does not help very much; see Derksen and Keselman (1992).
9. It allows us to not think about the problem.
10. It uses a lot of paper.

“All possible subsets” regression solves none of these problems.

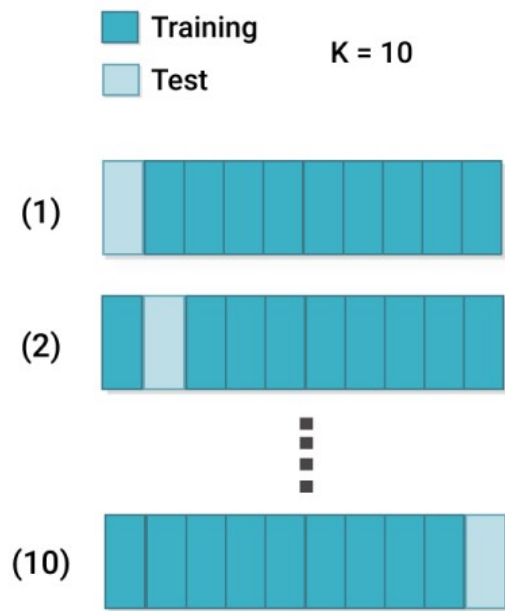
Current modern solution: **Shrikage**: Ridge and LASSO

$$RSS = \sum_{\vec{x}} (y_i - x_1 \cdot \beta_1 - x_2 \cdot \beta_2 - \beta_0)^2 \quad \rightarrow \quad \text{local optimum (estimate): } \min RSS \rightarrow \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_0$$

$$RSS \text{ and regularization} \quad \rightarrow \quad \text{local optimum (estim): } \min (RSS + Lasso) \quad \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_0$$

$$\text{regularization} = \lambda \sum \beta_i^2 (\text{Ridge}) \quad \text{or} \quad = \lambda \sum |\beta_i| (\text{Lasso}) \quad \text{Hyperparameter } \lambda!$$

- > **Ridge** (glmnet alpha = 0) typically only makes parameters small(~ 0), but has better performance
- > **LASSO** (glmnet alpha = 1) typically removes variables completely($=0$), but has a little bit worse performance
- > Standardization of variables needed (normalize intervals so that all X are same, e.g. Age 1-40, Height 120-190)
- > Learn lambda with Cross-Validation technique



Choose a set of lambdas: $\lambda \in [10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^0]$

1) Split train into 10 folds (or you can split inside for loop)

2) For every lambda λ :

For every possible validation folds:

2.1) $\min [\text{RSS} + \lambda * \text{regulariz}] = \text{fit on 9/10 train folds}$

2.2) Compute MSE = RSS on 1/10 test fold

3) Average MSE test errors from 2.2) for every lambda

4) Choose the best lambda by validation score from 3)

> K-fold for small subset (can run multiple times, too little dataset = use processing power)

> Hold-out for large subset (if my model or data are huge, just run once)

More details:

<https://medium.com/analytics-vidhya/regularization-and-cross-validation-how-to-choose-the-penalty-value-lambda-1217fa4351e5>

- Today a big homework will be presented and the countdown starts
 - You will have ~ 2 weeks to complete it, the time is parallel to every-week small HW

3	7.10.	JB, AA, JK	Shrunked linear regression	 san_fs.zip ,  assignment1.zip  lab3.zip  Linear regression.pptx
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